# SOC Lab Report - Week-1

## Security Operations Center — Threat Detection, Monitoring & Vulnerability Assessment

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## 1. Introduction

This report documents the hands-on implementation of a Security Operations Center (SOC) lab environment. The goal was to configure and demonstrate key security tools that a Tier 1 SOC Analyst would use in a realworld environment. Clear documentation with screenshots is provided for each tool to show working configurations and results.

The lab was designed to simulate real-world attack scenarios and demonstrate the complete detection and investigation chain — from network-level intrusion detection all the way through endpoint monitoring, centralized log management, and vulnerability assessment.

The objective was to simulate real-world SOC operations including detection, analysis, and response to security events.

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## 2. Tools Used

| Tool | Version | Purpose |
|---|---|---|
| Kali Linux | 2025.2 | Primary security operations VM — Snort, Wazuh, Osquery |
| Windows 11 Pro | 10.0.26100.4946 | Endpoint target — Wazuh agent, Osquery, Event Viewer |
| Metasploitable 2 | Ubuntu 8.04 / 10.04 | Intentionally vulnerable target machine for Nessus scan |
| Snort | 3.11.1.0 | Network IDS — real-time packet inspection and alerting |
| Osquery | Latest | Endpoint monitoring — SQL-based process and system queries |
| Wazuh | 4.7.5 | Open-source SIEM — log analysis, threat detection, alerting |
| Nessus Essentials | 10.11.3 / 19.16.3 | Vulnerability scanner — CVE detection and risk assessment |
| Oracle VirtualBox | Latest | Hypervisor managing all virtual machines |

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## 3. SOC Fundamentals and Operations

A Security Operations Center (SOC) is responsible for:

- Continuous monitoring of systems
- Detecting suspicious activities

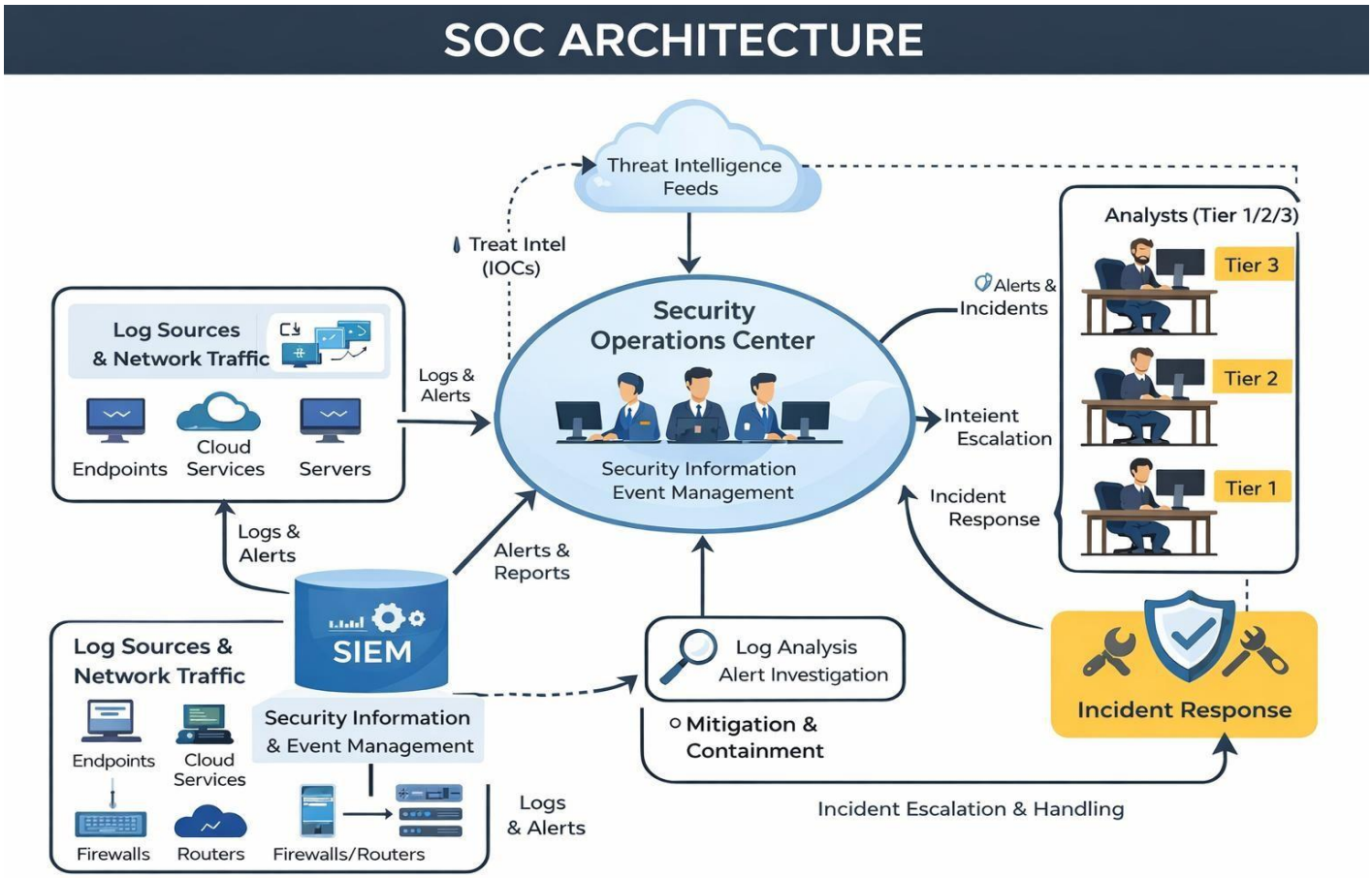
- Responding to incidents **SOC Roles**
- **Tier 1 Analyst** → Alert monitoring & triage
- **Tier 2 Analyst** → Deep investigation
- **Tier 3 Analyst** → Threat hunting & advanced analysis

• **SOC Manager** → Overall coordination **Key**

**Functions Performed in Lab**

- Log analysis using Wazuh
- Alert triage using dashboards
- Threat detection using Snort
- Vulnerability identification using Nessus

**SOC Architecture Diagram**




## 4. Snort – Intrusion Detection System (IDS)

Snort is an open-source network intrusion detection system that performs real-time traffic analysis and packet logging. In this lab, Snort 3 was installed on Kali Linux and configured with custom detection rules.

### 4.1 Custom Rule Configuration

Custom rules were written in /etc/snort/rules/local.rules to detect specific types of network traffic. The following rules were created:

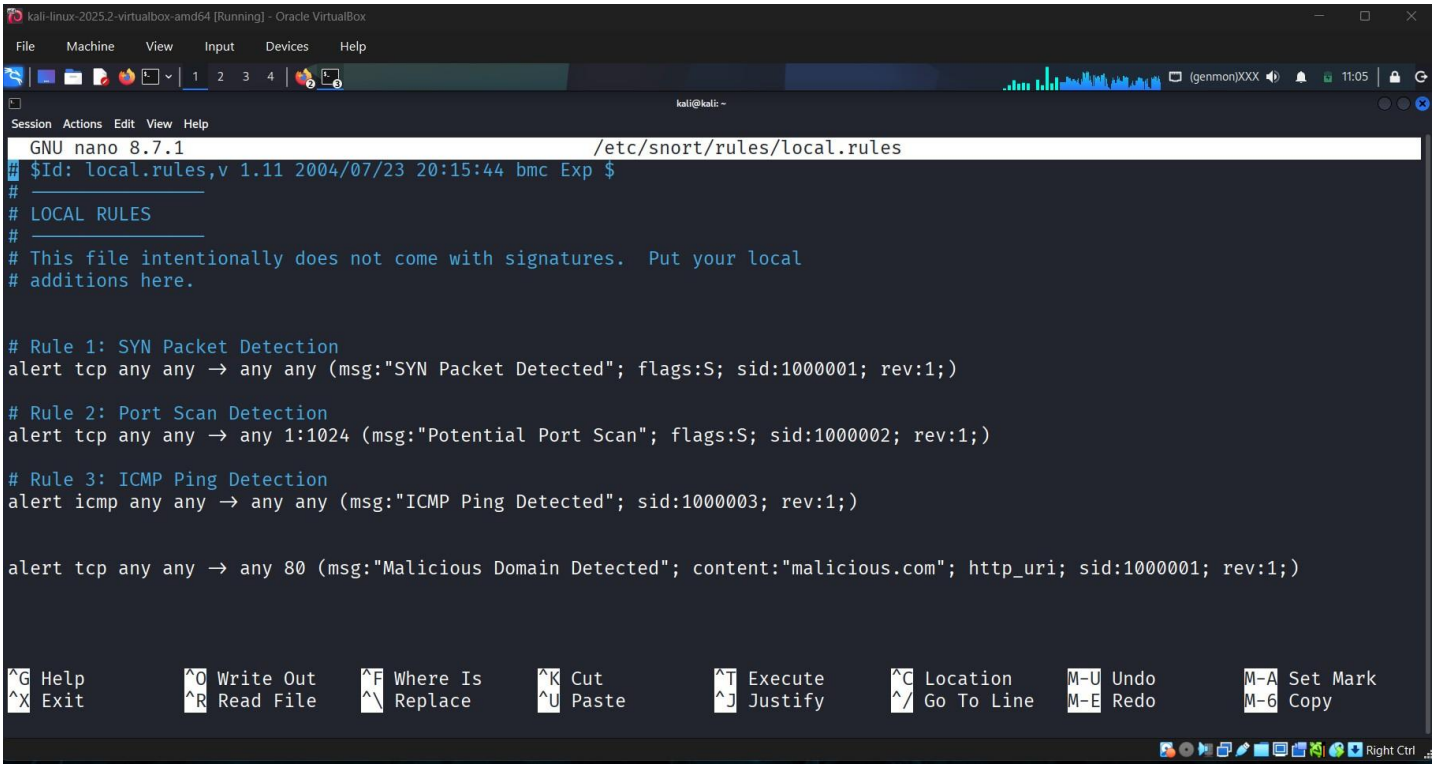



**Figure 1: Snort custom rules configured in /etc/snort/rules/local.rules**

## 4.2 Configuration Validation

After writing the rules, the Snort configuration was tested using the test (-T) flag to ensure there were no syntax errors. The command used was:

```
sudo snort -c /etc/snort/snort.lua -T
```

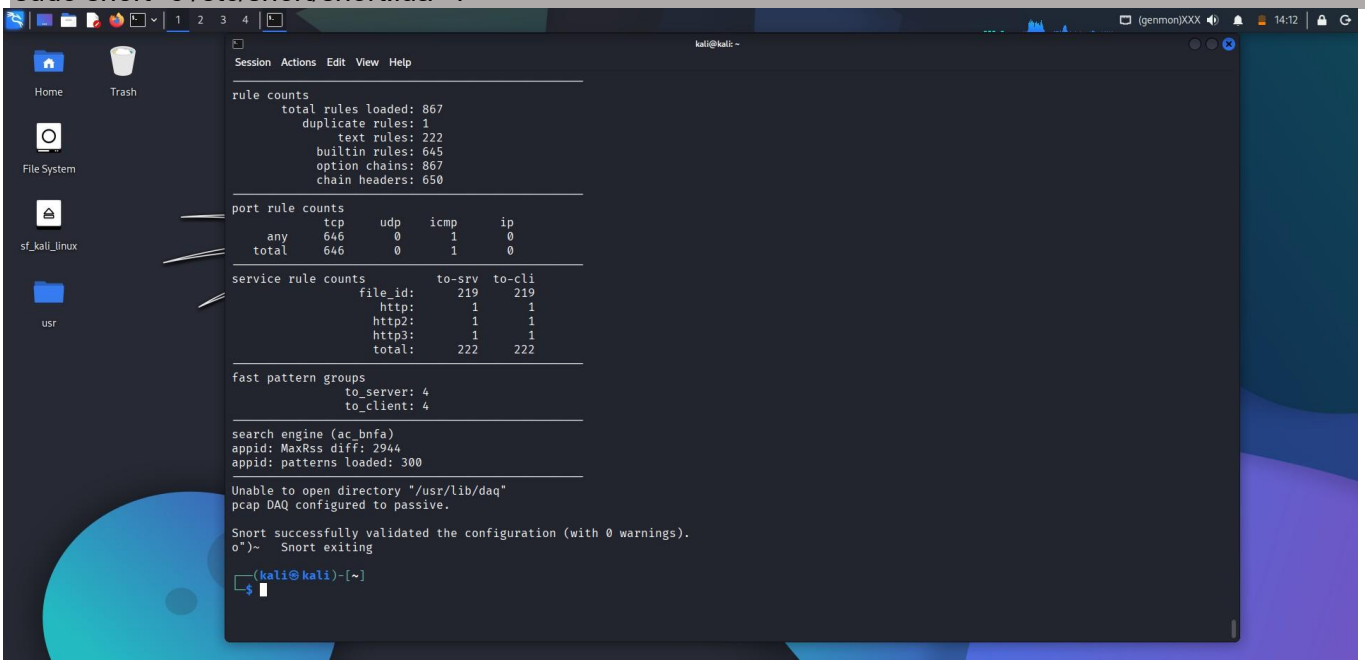



**Figure 2: Snort configuration validation showing 'successfully validated' with 0 warnings**

### 4.3 Live Alert Generation

Snort was run in live monitoring mode on interface eth1. HTTP requests were sent to malicious.com using curl to trigger the configured rules.

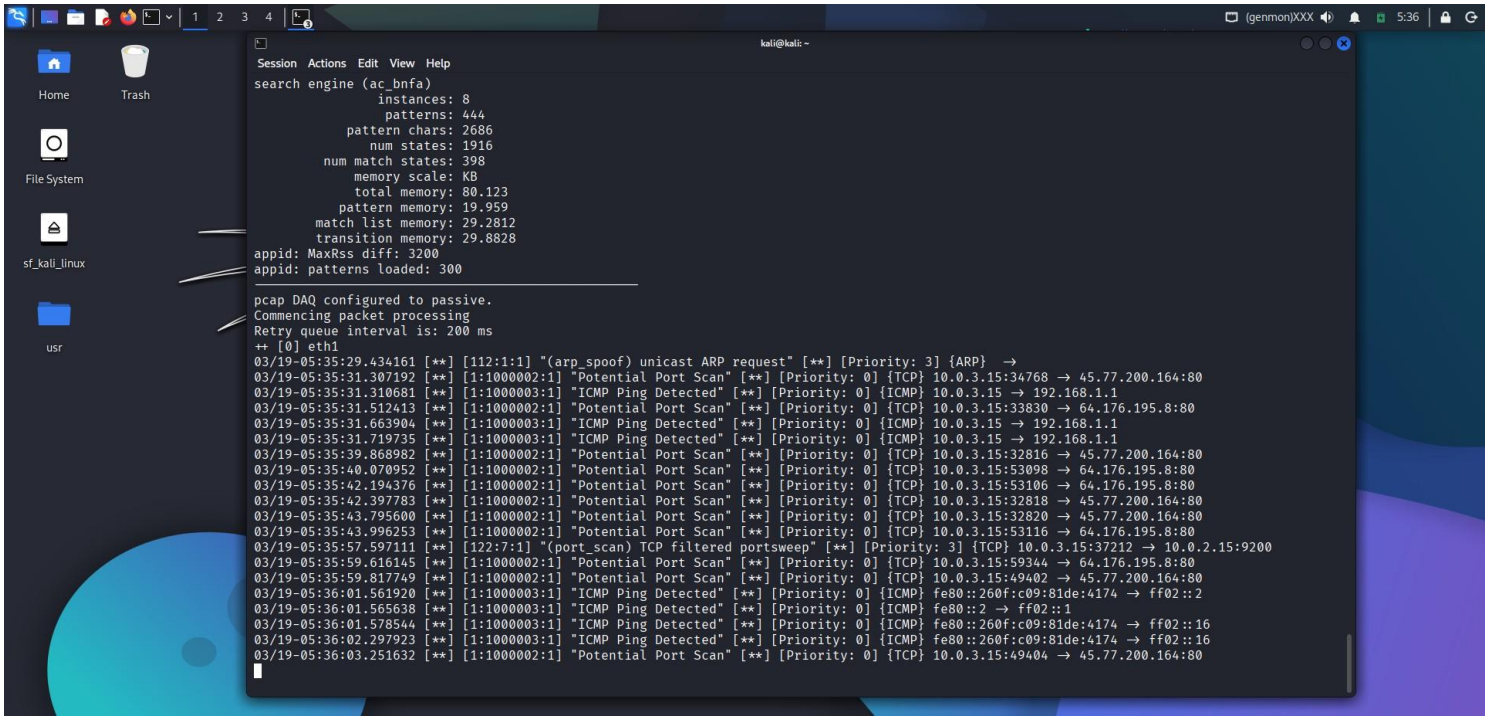



**Figure 3: Snort live alerts firing on eth1**

The following alert types were generated in real time:

- Unicast ARP request detected [Priority: 3]
- Multiple TCP SYN packets to external IPs [Priority: 0]
- Echo requests from 10.0.3.15 to 192.168.1.1
- Port sweep on 10.0.2.15:9200

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## 5. Osquery Endpoint Process Monitoring

### 5.1 Objective

Osquery exposes the operating system as a high-performance relational database, allowing analysts to write SQL queries to monitor running processes, network connections, user accounts, and more. It was installed on the Windows 11 VM to demonstrate endpoint visibility.

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### 5.2 Process Query Output

The osqueryi interactive shell was launched with Administrator privileges from C:\Program Files\osquery\. The following query was executed to retrieve all currently running processes:

```
SELECT pid, name, path FROM processes;
```

The query returned a full list of running processes including Chrome, osqueryd, svchost, msiexec, and various Windows system services — giving complete endpoint visibility at that moment in time.

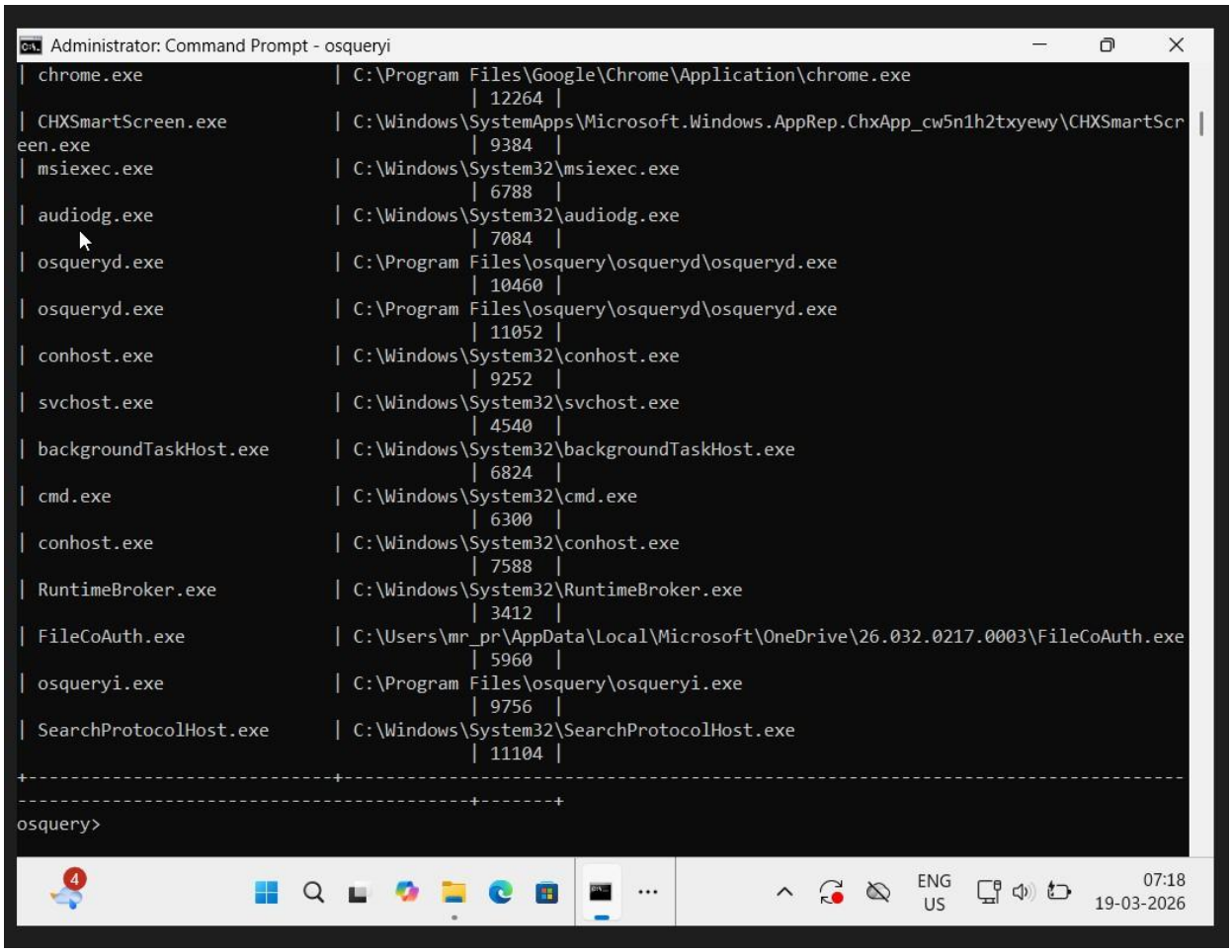


**Figure 4: Osquery on Windows VM — SELECT pid, name, path FROM processes**

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### 5.3 Malicious Process Simulation — MITRE ATT&CK T1059

To simulate malicious process execution, multiple CMD windows were opened simultaneously. The following targeted query was run to detect them:

```
SELECT pid, name, path FROM processes WHERE name LIKE '%cmd%';
```

The query returned 3 active cmd.exe processes (PIDs: 6300, 8372, 11196), confirming successful detection. This activity maps to MITRE ATT&CK Technique T1059 — Command and Scripting Interpreter, which covers adversary use of command-line interfaces to execute malicious commands.

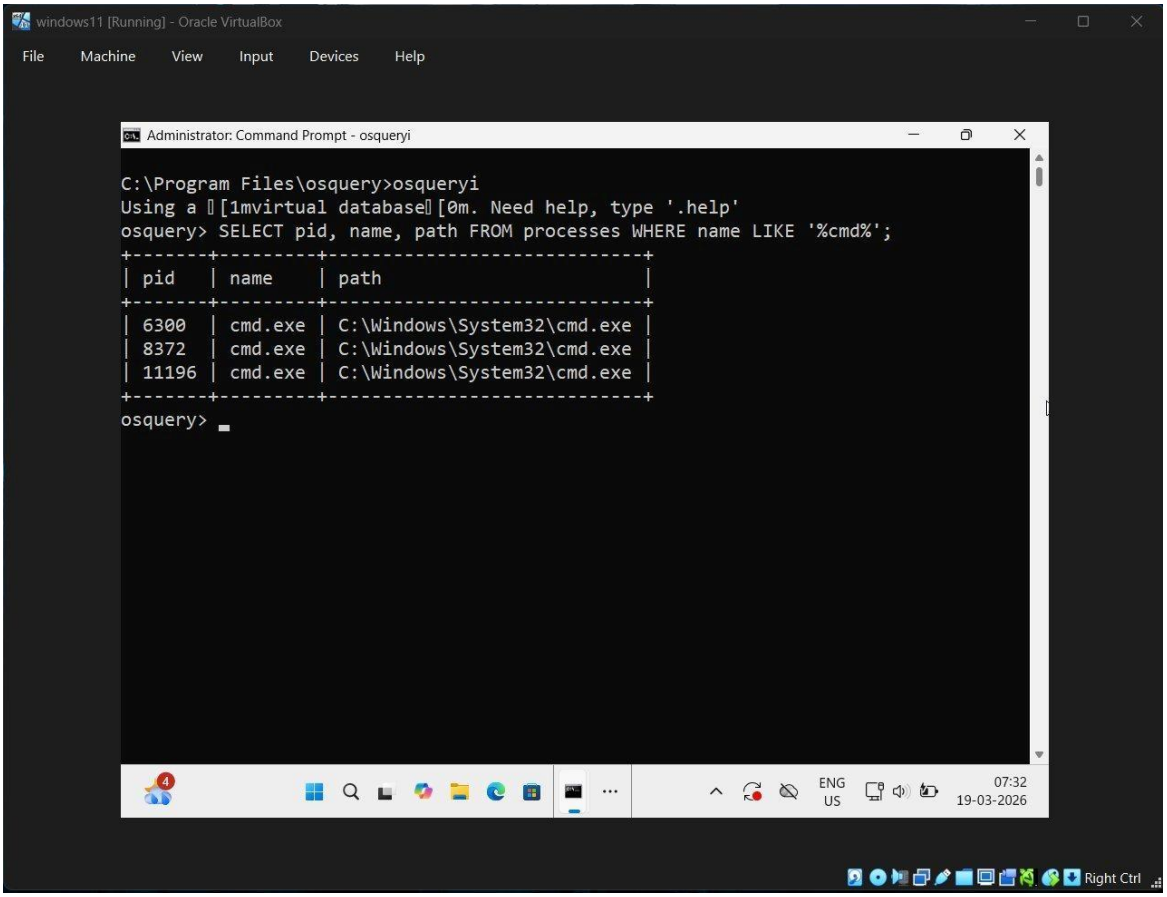


**Figure 5: Osquery detecting 3 cmd.exe processes (PIDs 6300, 8372, 11196) — MITRE ATT&CK T1059 simulation confirmed**

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## 6. Wazuh SIEM — Security Information & Event Management

### 6.1 Objective

Wazuh is an open-source security platform that combines SIEM, XDR, and SOAR capabilities. It was deployed on Kali Linux using Docker and configured to monitor both the Kali Linux system and the Windows 11 endpoint.

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### 6.2 Dashboard Overview

The Wazuh dashboard was accessible at https://127.0.0.1 after Docker deployment. It provides a unified interface for viewing connected agents, security events, compliance data, and threat intelligence — all in one place.

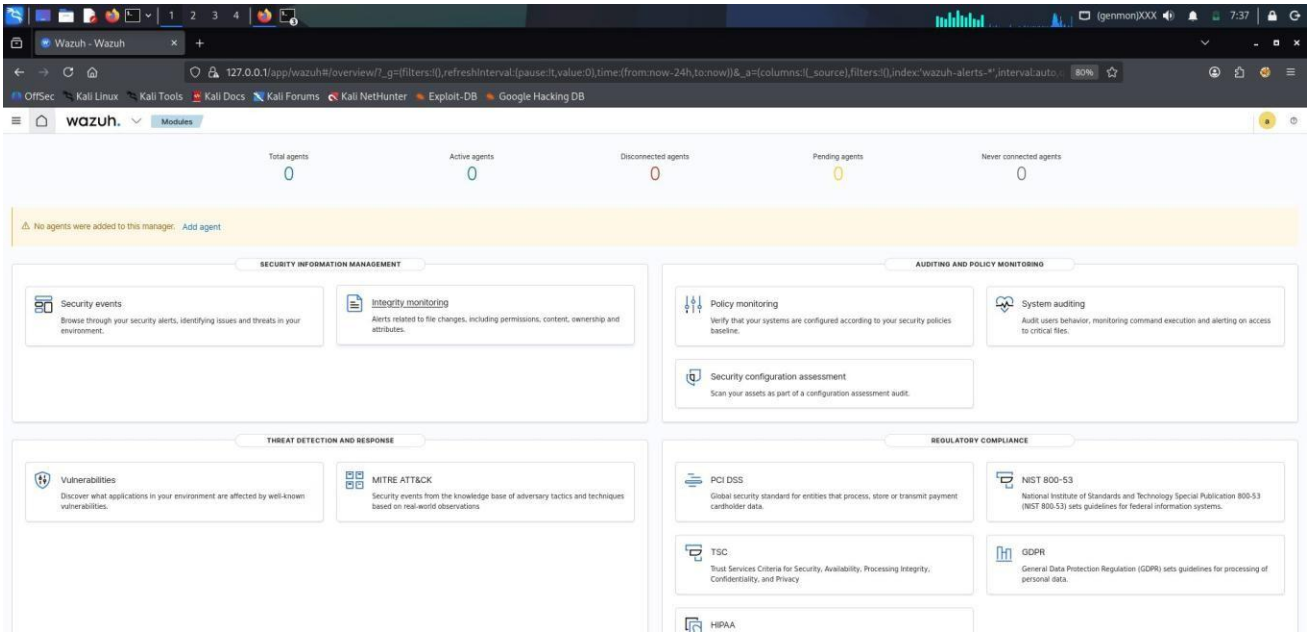


**Figure 6: Wazuh dashboard overview**

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### 6.3 Agent Connectivity

Two Wazuh agents were enrolled and connected to the Wazuh manager. Both showed Active status with 100% agent coverage:

| Agent ID | Name | IP Address | Operating System | Version | Status |
|---|---|---|---|---|---|
| 001 | kali-agent | 127.0.0.1 | Kali GNU/Linux 2026.1 | v4.7.5 | Active |
| 002 | windowagent | 192.168.56.105 | Windows 11 Pro 10.0.26100.4946 | v4.7.5 | Active |

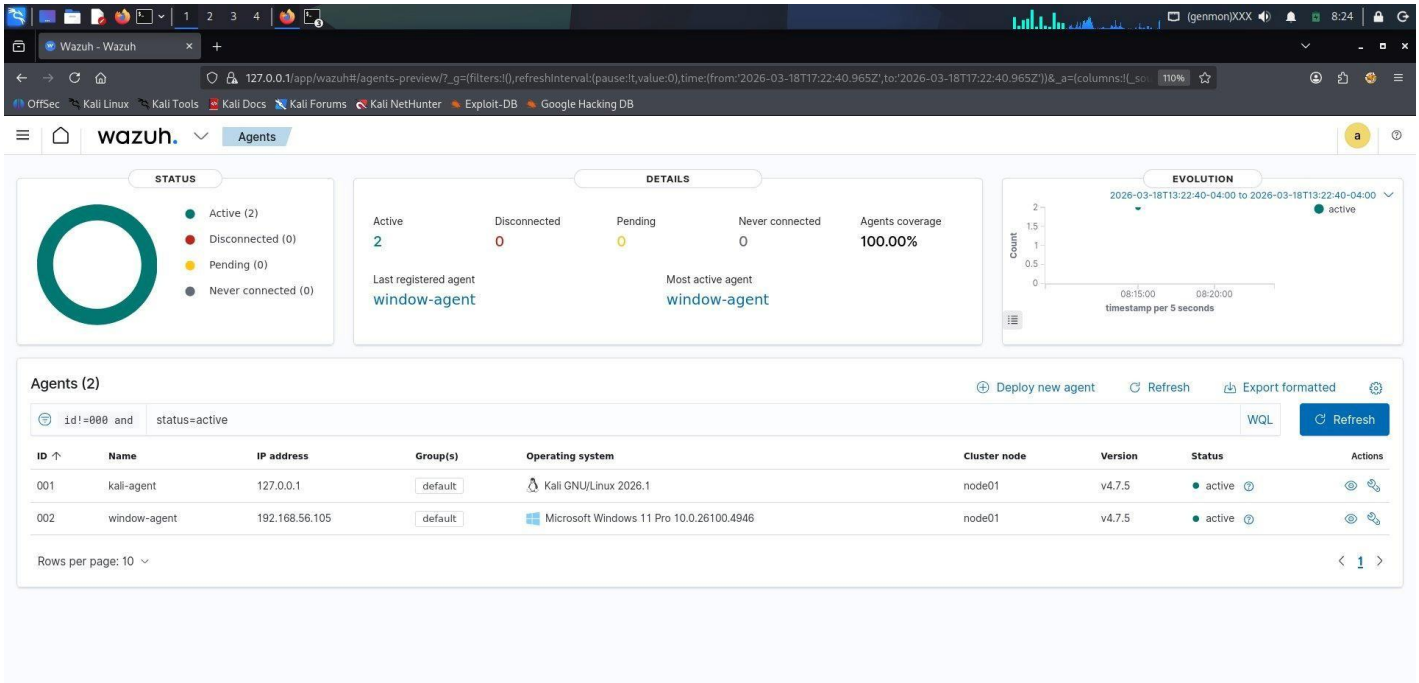


**Figure 7: Wazuh Agents page — kali-agent (001) and window-agent (002) both Active**

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## 6.4 Brute Force Detection

SSH brute force login attempts were simulated against the Windows agent from Kali Linux. Wazuh captured and displayed all authentication failures in real time. The Security Events dashboard for window-agent (002) showed:

- Total Events: 499
- Brute force attempts: 21
- Authentication Successes: 24
- Valid Accounts (Defense Evasion, Persistence, Privilege Escalation, Initial Access)

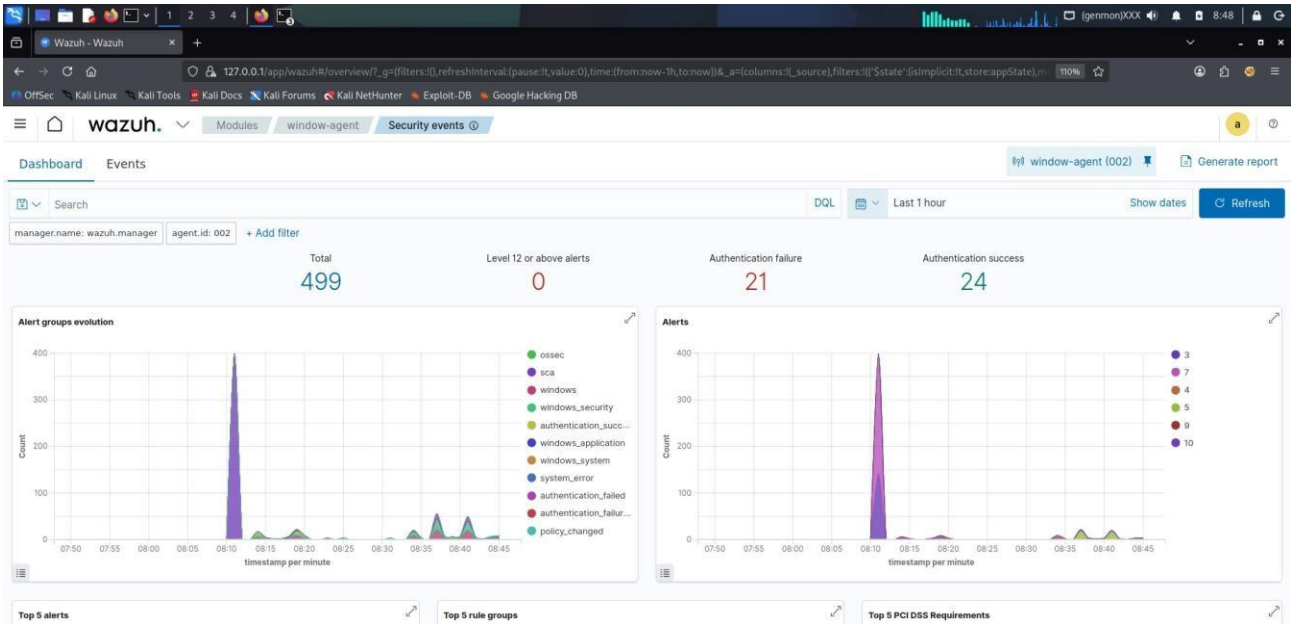


**Figure 8: Wazuh Security Events — 499 total events, 21 authentication failures detected**

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## 6.5 Custom Alert Rule & Security Alerts

A custom Wazuh rule was configured to detect repeated failed login attempts. The Security Alerts panel confirmed the rule was actively triggering. The MITRE ATT&CK framework was automatically applied to relevant alerts.

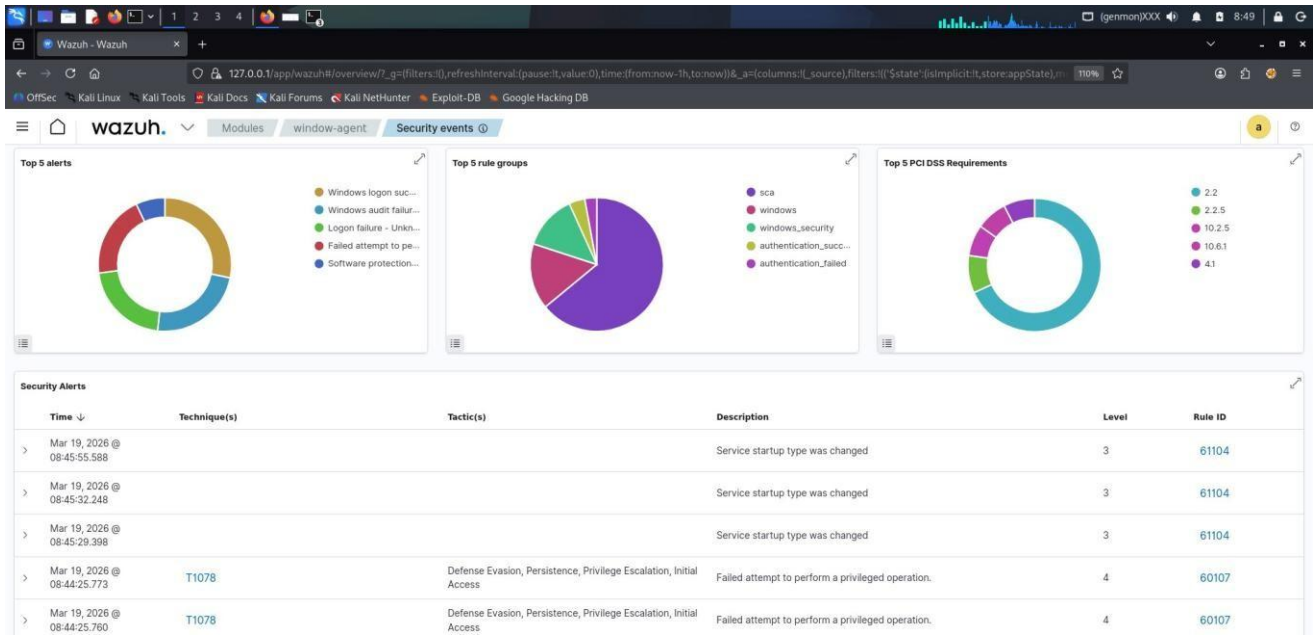


**Figure 9: Wazuh Security Alerts — MITRE ATT&CK T1078 mapped, Rule ID 60107 triggered**

## 7. Windows Event Viewer — Log Analysis

### 7.1 Objective

Windows Event Viewer is a built-in Windows tool for examining system, application, and security logs. It was used to investigate failed login attempts on the Windows 11 VM by filtering the Security log for Event ID 4625 — the standard Windows event for failed logon attempts.

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### 7.2 Event Forensics — Event ID 4625

Individual event details were examined to extract forensic information relevant to the brute force investigation. The event revealed the following:

| Field | Value | Forensic Significance |
|---|---|---|
| Event ID | 4625 | Failed logon attempt — standard Windows security event |
| Log Name | Security | Windows Security audit log |
| Timestamp | 19-03-2026 at 05:37:43 | Exact date and time of attack attempt |
| Workstation Name | KALI | Name of the attacking machine |
| Source Network Address | 192.168.56.102 | Kali Linux IP — confirmed attacker |
| Source Port | 46104 | Ephemeral port used by the attacker |
| Authentication Package | NTLM | Authentication protocol used |
| Keywords | Audit Failure | Confirms failed login — not a success |
| Target Computer | Dhruvil_003 | Victim Windows machine name |

The source IP 192.168.56.102 (Kali Linux) matched exactly with the Wazuh brute force alerts, confirming successful cross-tool event correlation. This demonstrates how Windows Event Viewer and a SIEM system work together to build a complete incident picture.

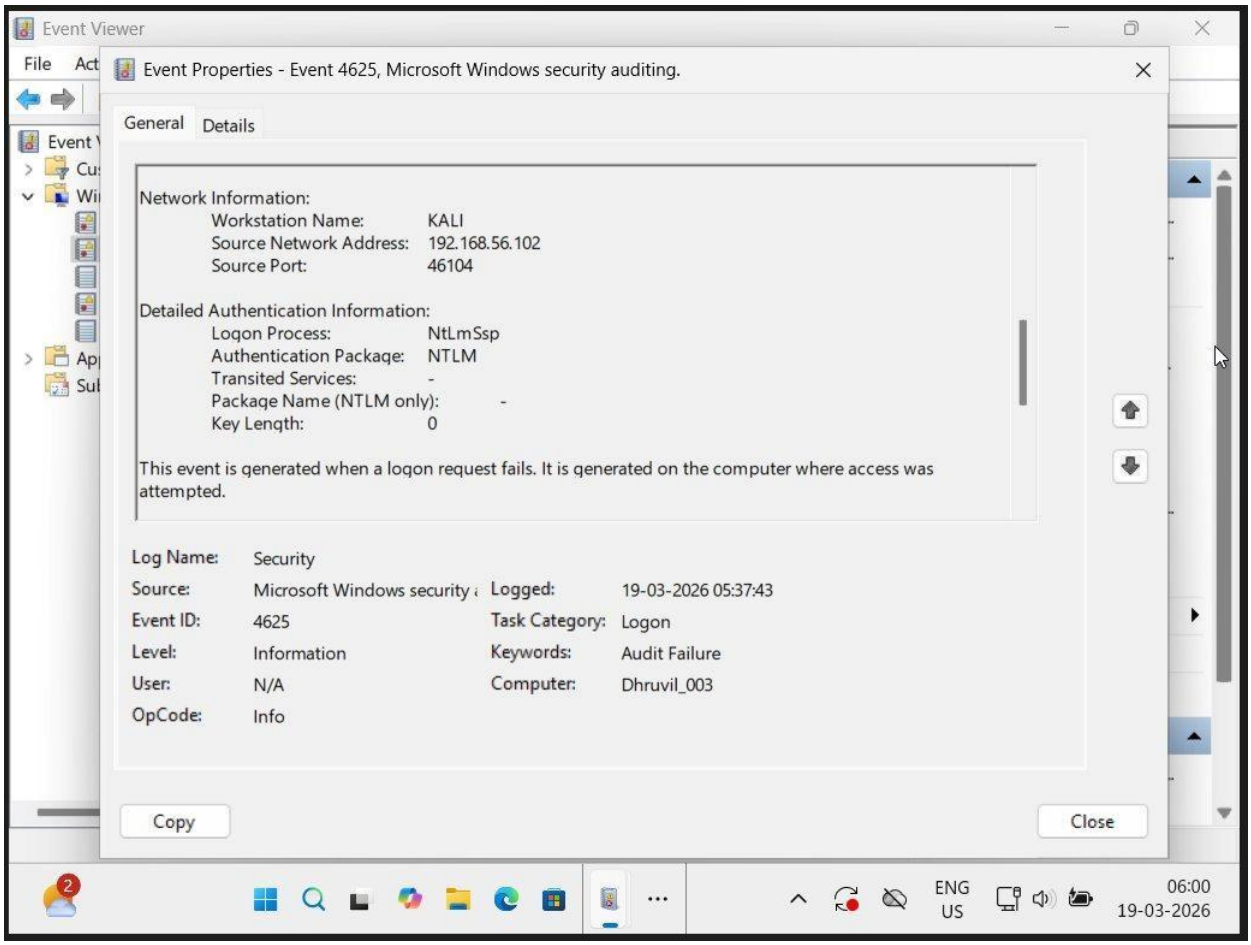


**Figure 10: Windows Event Viewer — Event ID 4625 detail showing source IP 192.168.56.102 (Kali), timestamp 05:37:43, and NTLM auth failure**

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## 8. Nessus — Vulnerability Assessment

### 8.1 Objective

Nessus Essentials is an industry-standard vulnerability scanner used to identify security weaknesses, misconfigurations, and known CVEs across networked systems. It was installed on the Windows 11 VM and used to scan the Metasploitable target machines.

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### 8.2 Scan Configuration

Two separate scans were configured and launched against Metasploitable targets:

- Target: 192.168.56.104, unauthenticated scan, Duration: 19 minutes, Result: 39 vulnerabilities

- Target: Metasploitable VM, SSH credentials (msfadmin / msfadmin), Duration: 24 minutes, Result: 91 vulnerabilities

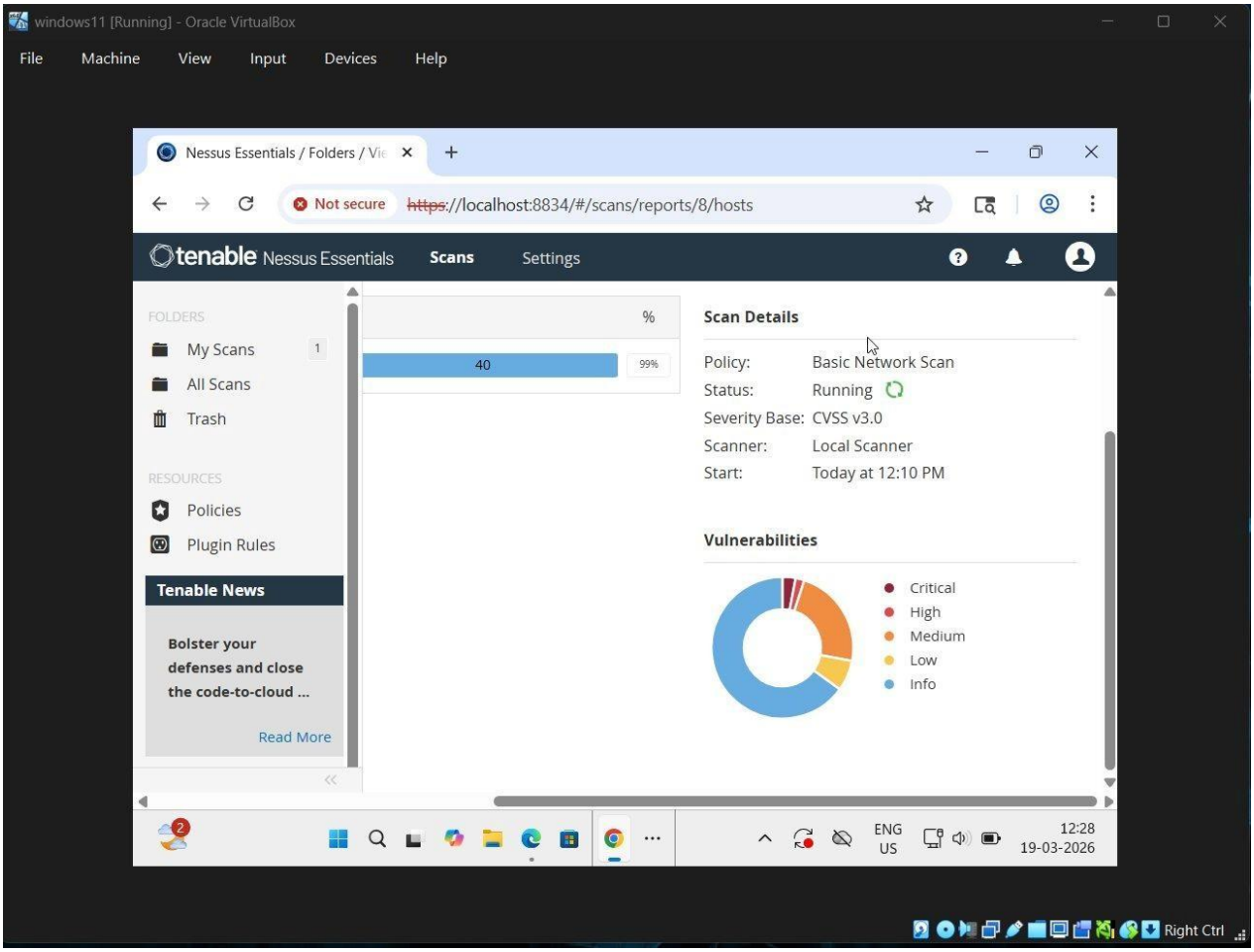


**Figure 11: Nessus Basic Network Scan in progress — scan running at 40% with vulnerability chart loading,target visible**

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### 8.3 Top Vulnerabilities Found (CVSS Sorted)

The credential scan returned 91 total vulnerabilities. The most critical findings, sorted by CVSS score, are shown below:

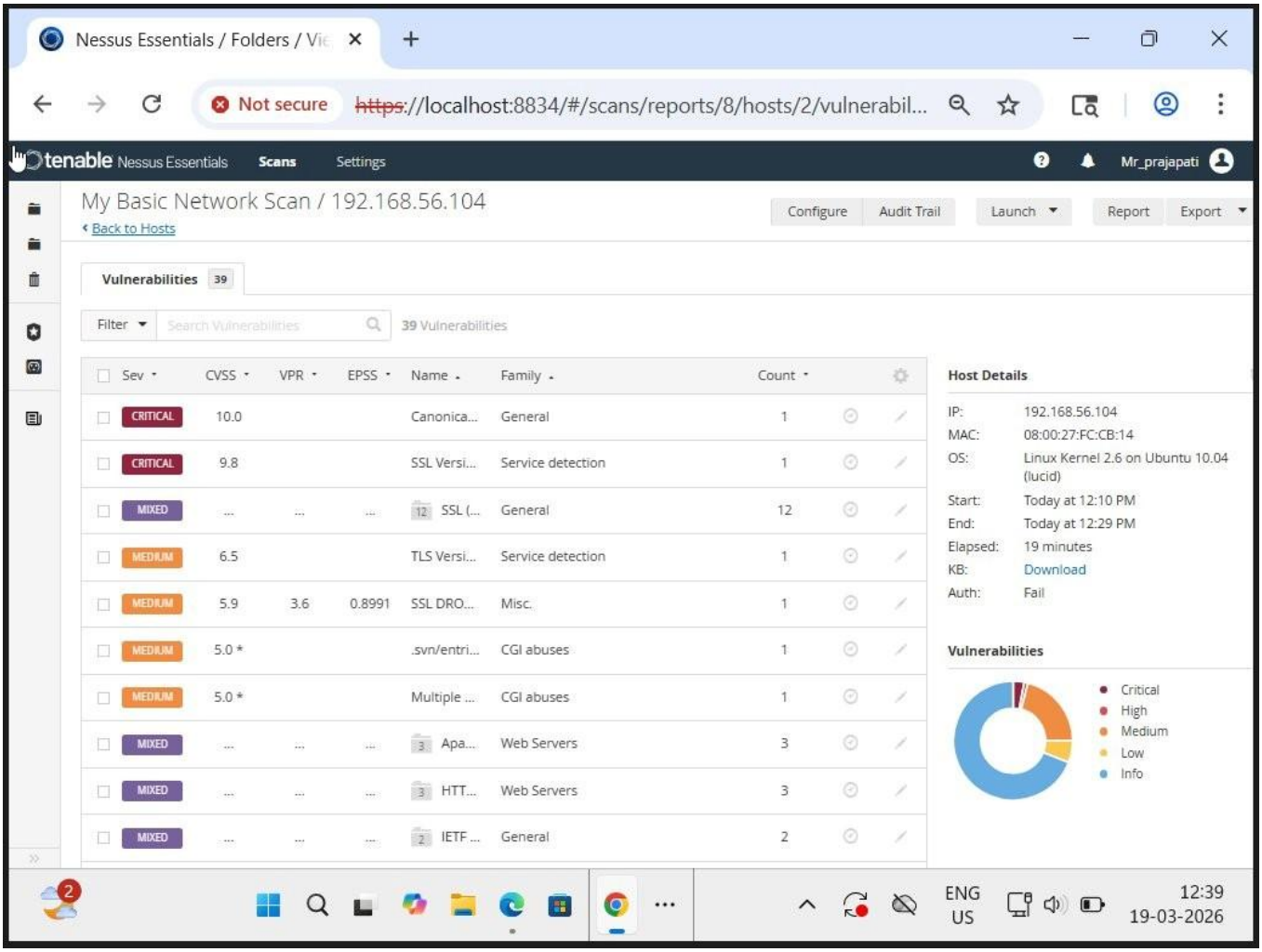


**Figure 12: Nessus credential scan results — CRITICAL 10.0 (VNC), CRITICAL 9.8 (Bash ShellShock, Weak Keys) as top findings**

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### 8.4 Report Export

After scan completion, the vulnerability report was exported for documentation. Since Nessus Essentials does not include the built-in PDF export feature (available only in paid tiers), the browser's print-to-PDF function was used to capture the complete vulnerability list as a PDF document.

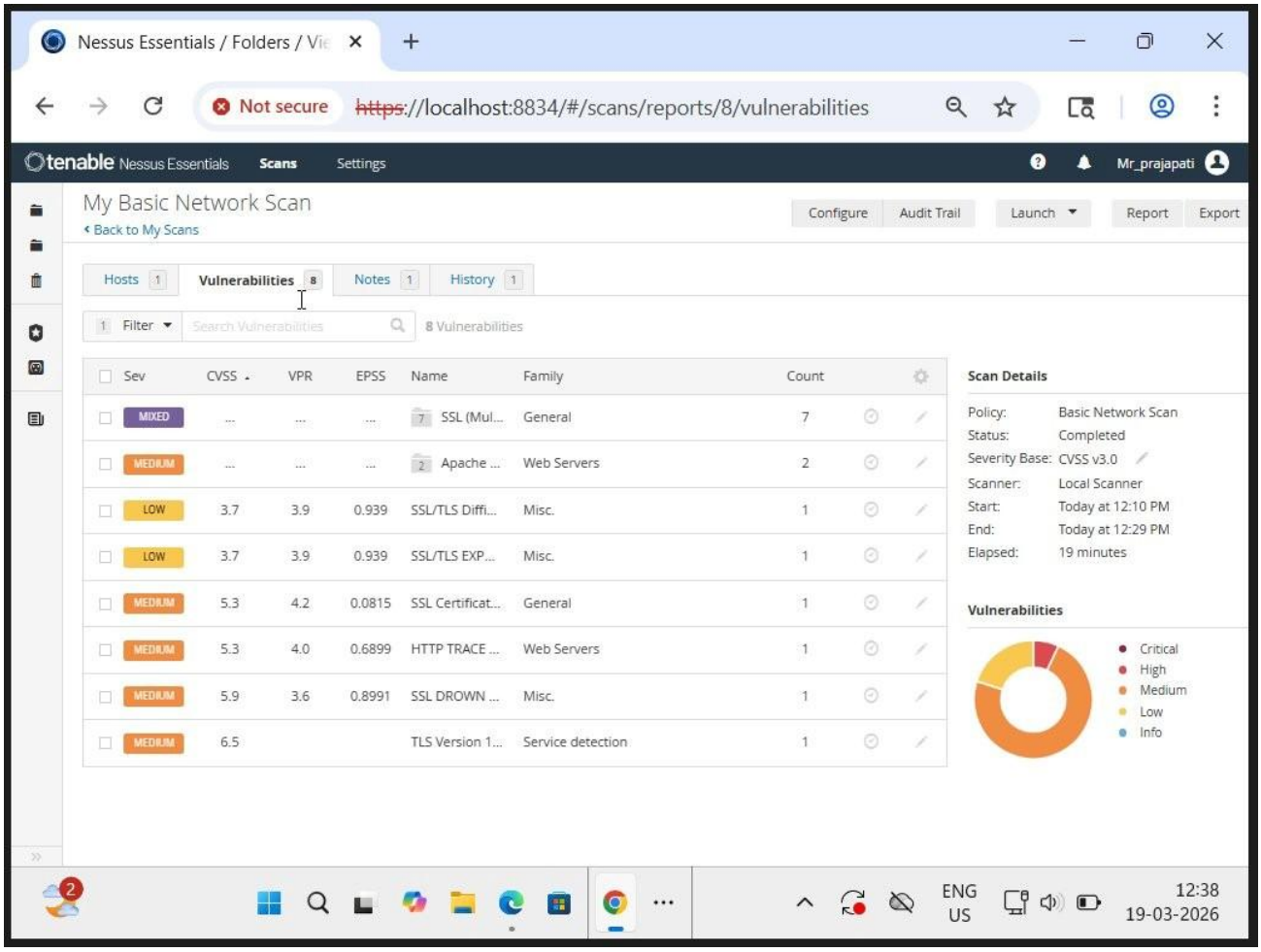


**Figure 13: Nessus vulnerability results page — exported via browser print-to-PDF confirming scan completion and findings**

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## 9. Security Analysis

The lab confirmed several critical security gaps that are common in real-world environments:

- The Metasploitable VNC server used no password, and the SSH service accepted msfadmin:msfadmin — credentials cracked immediately. Fix: Enforce strong unique passwords, disable VNC where not required, implement SSH key-based authentication only.
- The SSH service accepted unlimited failed login attempts without any rate limiting or lockout. Fix: Configure MaxAuthTries 3 in /etc/ssh/sshd_config and deploy fail2ban to automatically block repeated failed logins.
- SSL v2/v3 and TLS 1.0 were active on multiple services — protocols with known vulnerabilities. Fix: Disable legacy SSL/TLS in server configurations and enforce TLS 1.2 or higher.
- The Wazuh dashboard detected all 21 authentication failures in real time and automatically mapped the attack to MITRE ATT&CK T1078. This demonstrates how a properly configured SIEM significantly reduces the time to detect (MTTD) credential attacks.

- The Bash vulnerability was present and exploitable. Fix: Update Bash to a patched version immediately and conduct regular vulnerability scans to identify unpatched systems.

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## 10. Key Learnings

Building this multi-tool SOC environment provided hands-on experience with the complete security operations workflow. Configuring Snort, Wazuh, Osquery, and Nessus together highlighted how different tools cover different layers of visibility — network, endpoint, centralized logging, and vulnerability management — and how they work together to build a complete threat detection picture.

The brute force simulation against the Windows agent demonstrated how quickly weak credentials can be exploited and how a properly configured SIEM catches it. The MITRE ATT&CK mapping (T1059 for command execution, T1078 for credential abuse) confirmed that Wazuh's built-in detection logic aligns with real-world adversary tactics.

Correlating the Wazuh brute force alerts with the Windows Event Viewer Event ID 4625 records — both pointing to the same source IP (192.168.56.102) — demonstrated how multi-source log correlation works in practice. This is a core Tier 1 SOC analyst skill: confirming an incident across multiple independent data sources before escalating.

The Nessus credential scan against Metasploitable revealed 91 vulnerabilities including ShellShock, VNC password exposure, and active backdoors — results that would be impossible to find with an unauthenticated scan alone. This reinforces the importance of authenticated scanning in a real vulnerability management program.

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## 11. Challenges Faced

- Nessus consumed 2.9GB RAM on Kali Linux and filled the disk with large log files (536MB each). Resolved by uninstalling Nessus from Kali and reinstalling on the Windows 11 VM which had sufficient available memory.
- After installing Osquery on Windows, the osqueryi command was not recognized in CMD because the install directory was not added to the system PATH. Resolved by navigating directly to C:\Program Files\osquery\ and running osqueryi from there.
- On fresh installation, Nessus required 20-30 minutes to compile plugins before scans could be launched. The New Scan button was greyed out and unclickable during this process. Resolved by waiting for compilation to complete.

- Initial validation used the wrong config path for Snort 3 (/etc/snort/snort.conf does not exist in v3). Resolved by locating the correct snort.lua configuration file path.

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## 12. Conclusion

This task provided comprehensive hands-on experience across all core Security Operations Center competencies. The lab successfully demonstrated the complete threat detection and investigation chain — from initial network packet inspection through endpoint monitoring, centralized SIEM correlation, Windows log forensics, and vulnerability assessment.

All 14 required screenshots were captured, covering every stage of the SOC workflow. The tools worked together exactly as they would in a real SOC environment: Snort detected suspicious network traffic, Osquery confirmed process-level activity, Wazuh correlated events and fired MITRE-mapped alerts, Event Viewer provided forensic evidence, and Nessus identified the vulnerabilities that made the attacks possible.

The most important takeaway from this lab is that no single tool provides complete visibility. Effective security operations require multiple overlapping layers — network, endpoint, and log analysis — correlated through a central SIEM platform. This is exactly what the lab environment demonstrated, and it reflects the architecture of real enterprise SOC deployments.

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## 12. References


Snort 3 Documentation: https://www.snort.org/documents

Osquery Documentation: https://osquery.readthedocs.io

Wazuh Official Documentation: https://documentation.wazuh.com

Tenable Nessus Essentials: https://www.tenable.com/products/nessus/nessus-essentials

MITRE ATT&CK Framework: https://attack.mitre.org

NIST SP 800-61 — Computer Security Incident Handling Guide: https://csrc.nist.gov/

CVE-2014-6271 (ShellShock): https://nvd.nist.gov/vuln/detail/CVE-2014-6271